



**MACQUARIE**  
University  
**BUSINESS SCHOOL**  
SYDNEY • AUSTRALIA

# Techniques in **B**usiness **A**alytics

**BUSA8000**

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## **Abstract**

### **Abstract**

This report presents a deep dive into LuminaTech's sales and customer data from 2012 to 2013, transforming raw information into clear, actionable insights to support the company's strategic goals. Using a blend of exploratory data analysis, statistical testing, and predictive modeling, we uncover key patterns that shape sales performance and customer behavior. For instance, contrary to expectations, new customers make larger purchases on average than returning ones—a finding that shifts how loyalty programs could be structured. To aid in operational planning, we develop a SARIMA model that forecasts 2014 sales trends, highlighting expected peaks and troughs, which can guide inventory and staffing adjustments for optimal readiness. Additionally, our customer churn analysis points to critical mid-year declines and regional differences, suggesting targeted retention strategies that could mitigate potential revenue loss. This report not only underscores the value of data-driven insights but also illustrates how analytics can empower LuminaTech to enhance efficiency, build stronger customer relationships, and navigate market changes with confidence.

## **Acknowledgement**

Acknowledgement Statement by students:

We acknowledge that we have only used GAITs (e.g., ChatGPT) in drafting and proofreading this assignment, which is permitted in the assignment instructions.



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## Introduction

This report, prepared by the group(us) offers a comprehensive analysis of company data to guide strategic decisions in sales performance, customer retention, and inventory management. The dataset was finalized by merging 2012 and 2013 dataset that includes fiscal and calendar year data, customer and product codes, pricing adjustments, and order and invoice records. Through strategic data cleaning, preparation and analysis, the report delivers actionable insights to enhance operations and support growth.

**Section 1** details the data cleaning process, addressing missing values, standardizing categorical data, and managing outliers to ensure accuracy and reliability. Missing data was handled selectively to preserve critical information, and outliers were adjusted with tailored approaches, particularly for key metrics like sales volume and order quantities.

**Section 2** presents five key insights, exploring trends in sales performance, customer retention, and regional profitability. Each insight is paired with visualizations and recommendations for decision-making. For instance, analysis of sales trends identified peak periods, supporting staffing and marketing planning, while regional profitability insights reveal high-performing areas for focused promotional efforts.

**Section 3** applies hypothesis testing to examine differences in profit margins across fiscal years and compare order quantities between new and returning customers. These statistically significant insights support customer segmentation and targeted marketing strategies.

In **Section 4**, regression analysis assesses how pricing adjustments, fiscal month, and customer district influence sales and order quantities, clarifying the impact of key factors on LuminaTech's metrics.

**Section 5** introduces a SARIMA model that predicts sales for 2014, helping LuminaTech prepare for anticipated mid-year sales peaks. This model's projections can support inventory planning and resource allocation, ensuring readiness for both high-demand and quieter periods.

**Section 6** examines customer churn, identifying factors linked to attrition and recommending retention strategies for high-risk segments. Regional churn analysis, for example, pinpoints areas where retention efforts could be strengthened.

Overall, this report translates raw data into a strategic framework, providing insights that align with LuminaTech's goals. By equipping decision-makers with data-driven strategies, the report highlights the potential of analytics to optimize resources, boost operational efficiency, and support sustained growth across LuminaTech's business units.



## **Section 1 – Exploratory Data Analysis**

### **1.1 Data Import, Initial Exploration and Overview**

We began by importing the necessary libraries, including Pandas and NumPy, to facilitate data handling and manipulation. Two datasets, one from 2012 and another from 2013, were available for analysis. We merged these datasets to create a unified data frame, simplifying exploration and analysis by consolidating the data. To understand the data, we used the `.head()` function to display the first few rows, which revealed the dataset's structure and confirmed it includes 41 columns. Key columns, such as `value_sales`, `value_quantity`, and `value_cost`, stood out as potentially important for deeper analysis. We then used the `.shape` attribute to confirm that the dataset consists of 41 columns and a substantial 1,988,382 rows, indicating a large volume of data to work with.

Next, the `.info()` method provided a comprehensive summary, displaying data types for each column and highlighting any missing values. Finally, we used `.describe()` to review descriptive statistics, including mean, median, minimum, maximum, and standard deviation. Among the numerous columns, `value_sales`, `value_quantity`, and `value_cost` appeared most relevant for further analysis. These will likely serve as focal points in the subsequent steps.

### **1.2 Standardizing Categorical Data**

We checked the unique values in the `currency` and `business_area_code` columns to ensure consistency. In the `currency` column, we found an entry labeled 'AUS', which we assumed to be an error, so we used the `.replace()` function to standardize it to 'AUD'. In the `business_area_code` column, there were codes like '970', '920', '999', and '930' that represented the same business areas according to company standards. We converted these codes to 'LMP', 'FLD', 'OTH', and 'RWY', respectively, to align with the company's standardized naming conventions. This step helped maintain data consistency and accuracy across these columns, simplifying analysis.

### **1.3 Converting Data Types and Checking for Duplicates**

To ensure each column was set up correctly for analysis, we first checked the data types. The `accounting_date`, `order_date`, and `invoice_date` columns were initially stored as integers, so we converted them to datetime format to make date-related analysis easier and more accurate.

After that, I looked for any duplicate entries in the dataset. we found 8,648 duplicate rows, which we then removed to keep the dataset clean and focused. This step helps ensure that the analysis reflects unique and accurate data points.



### **1.4 Handling Missing Values and Dropping columns**

We began by examining the dataset for any missing values and identified several columns with null entries. The `environment_group_code` column had 162 missing values, which was a relatively small portion of the data. Additionally, the `currency` column had only 2 missing values. To handle these, We decided to drop the rows with missing entries in these columns, as this minimal data loss wouldn't significantly impact the overall dataset. Next, we looked at the `item_source_class` column, which was entirely empty, containing null values for all rows. Given that it lacked any useful data, we chose to remove this column completely to free up memory and streamline the dataset.

The `technology_group_code` column was a different case, with about 10% of its values missing. Dropping this column would have resulted in a significant loss of data, so we decided to retain it. The missing entries in this column generally represented items in an "na" group, which aligned with company practices. Keeping this column as-is allowed us to maintain data integrity without sacrificing a large amount of potentially valuable information.

Lastly, we noticed that the `dss_update_time` column contained the same value throughout the dataset. Since it didn't contribute any meaningful variation, we removed it to reduce memory usage and keep the dataset focused. This approach ensured that only necessary columns and rows remained, making the dataset cleaner and more manageable for analysis.

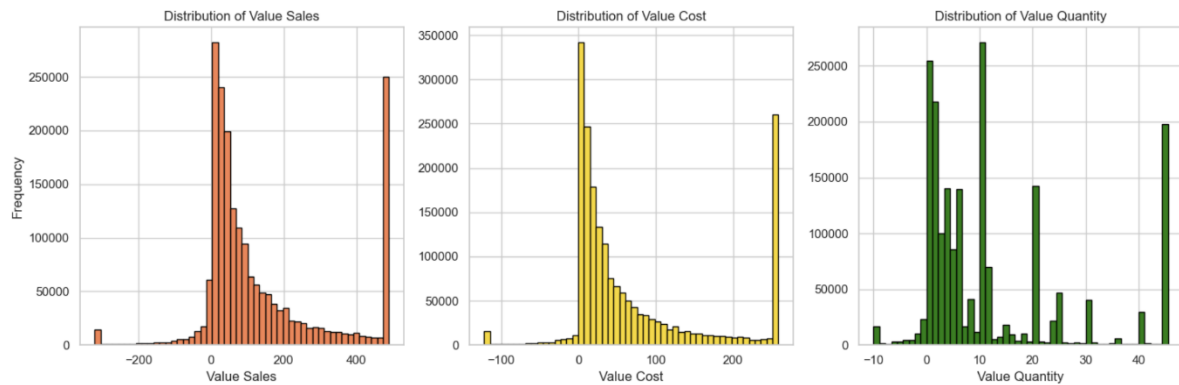
### **1.5 Identifying and Handling Outliers**

We began by creating box plots for `value_sales`, `value_cost`, and `value_quantity` to visually assess any outliers. Using the Interquartile Range (IQR) method, we found a considerable number of outliers: 272,907 in `value_sales`, 282,932 in `value_cost`, and 200,929 in `value_quantity`. These outliers fell well outside the typical range, and we needed a careful approach to handle them effectively.

To manage these outliers, we split each column's data into positive and negative groups, recognizing that each group might require different handling. The negative values in each column likely represent refunds or returns, so we wanted to retain these records rather than cap or modify them. Instead, we dealt with the positive and negative values separately, ensuring we kept the integrity of data related to refunds or returns. For positive values, we capped the outliers using the IQR limits, applying the calculated thresholds specifically to the positive group. This approach allowed us to adjust only the extreme positive values while preserving the original structure of the negative (refund-related) data. By addressing outliers in this targeted way, we minimized their impact on our dataset, keeping it robust for analysis while maintaining all critical data points.



## 1.6 Histograms



The histograms for `value_sales`, `value_cost`, and `value_quantity` reveal distinct distributions with both positive and negative values. **Value Sales:** The data is skewed, with most values clustered near zero and a long tail on the positive side. Negative values, likely representing refunds or returns, appear less frequently but are important for understanding customer behavior and transaction reversals.

**Value Cost:** Similar to `value_sales`, this distribution shows a concentration around zero with some high positive outliers, indicating costly transactions. Negative values, possibly cost adjustments or reversals, provide additional business insights. **Value Quantity:** This distribution has unique peaks, suggesting standard batch sizes or common order volumes. Both positive and negative extremes highlight unusual quantities, which may include large orders or returns.

## 1.7 Checking and Reducing Skewness

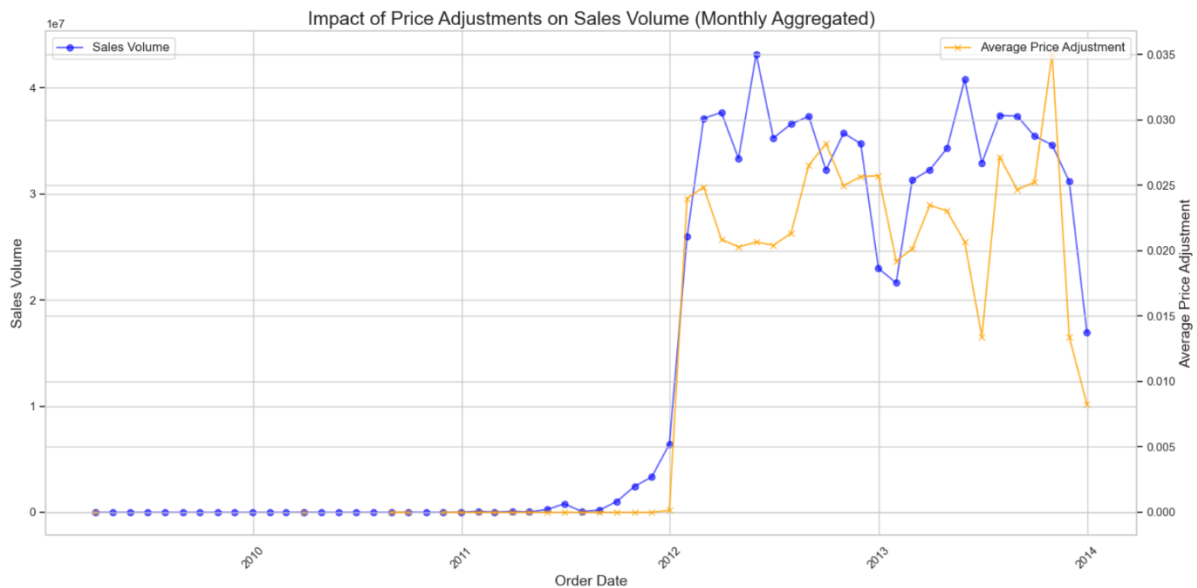
To understand the balance of our data, we checked the skewness in the `value_sales`, `value_cost`, and `value_quantity` columns. Initially, we found the data was skewed: **`value_sales`: 0.94, `value_cost`: 1.10, `value_quantity`: 1.33**. To address this, we used a **cube root transformation**. We chose this method because it works well for data with both positive and negative values, like our sales and returns, ensuring we didn't lose any critical information. After the transformation, skewness improved significantly: **`value_sales`: 0.94, `value_cost`: 0.27, `value_quantity`: 0.22**

This adjustment brought the data closer to a balanced distribution, making it more suitable for analysis and enhancing the accuracy of our future modeling efforts.



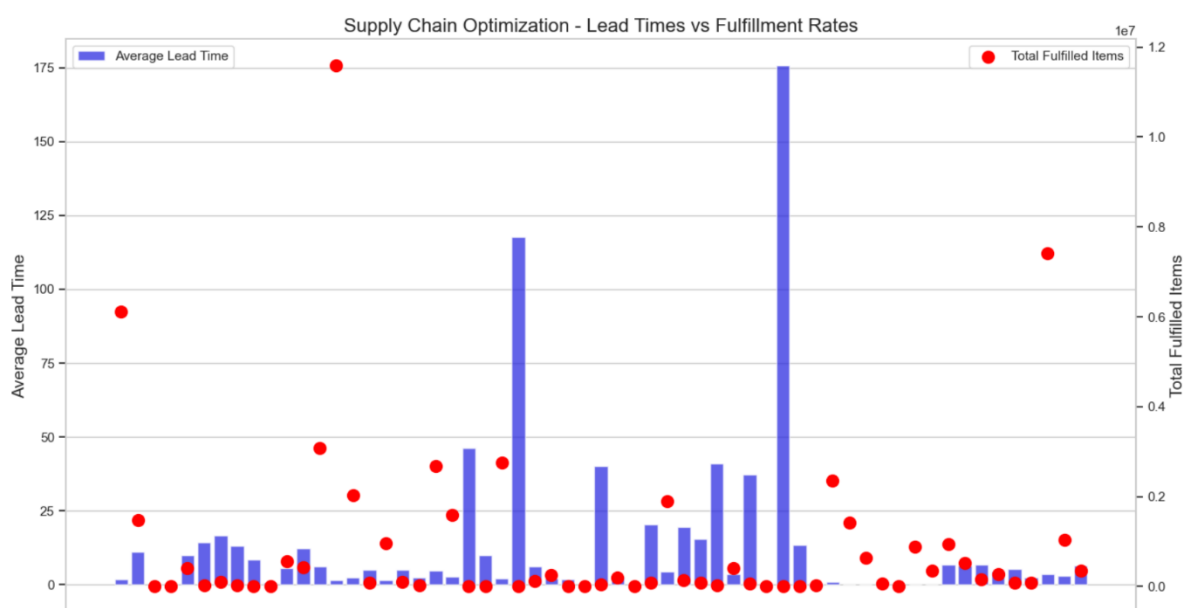
## Section 2: Exploratory Insights

### 2.1 Dynamic Pricing Impact Analysis



The line graph titled "Impact of Price Adjustments on Sales Volume (Monthly Aggregated)" shows the relationship between sales volume and average price adjustment over time from 2009 to 2014. The blue line with circular markers represents sales volume, while the orange line with cross markers represents average price adjustment. Starting around 2012, there's a notable increase in sales volume, with fluctuations in both sales volume and price adjustments thereafter. This visualization helps illustrate how price adjustments impact sales volumes over time, offering insights into the correlation between pricing strategies and sales performance.

### 2.2 Supply Chain Optimization Visualization

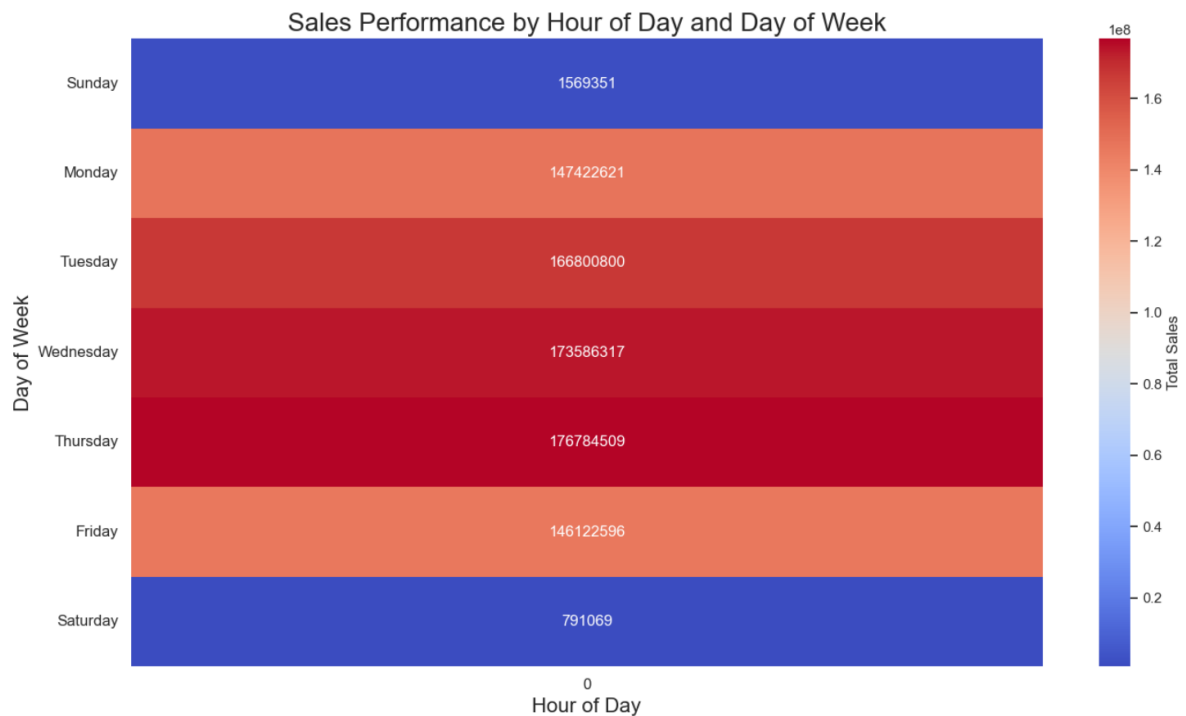




The above graph "Supply Chain Optimization - Lead Times vs Fulfillment Rates"

compares average lead times (blue bars) and total fulfilled items (red dots). The x-axis shows different data points, the left y-axis shows average lead times, and the right y-axis shows total fulfilled items. The graph highlights varying lead times, with some as high as 175, and fulfillment rates reaching up to  $1.2e7$ . This visualization illustrates the crucial relationship between lead times and fulfillment rates in optimizing supply chain processes.

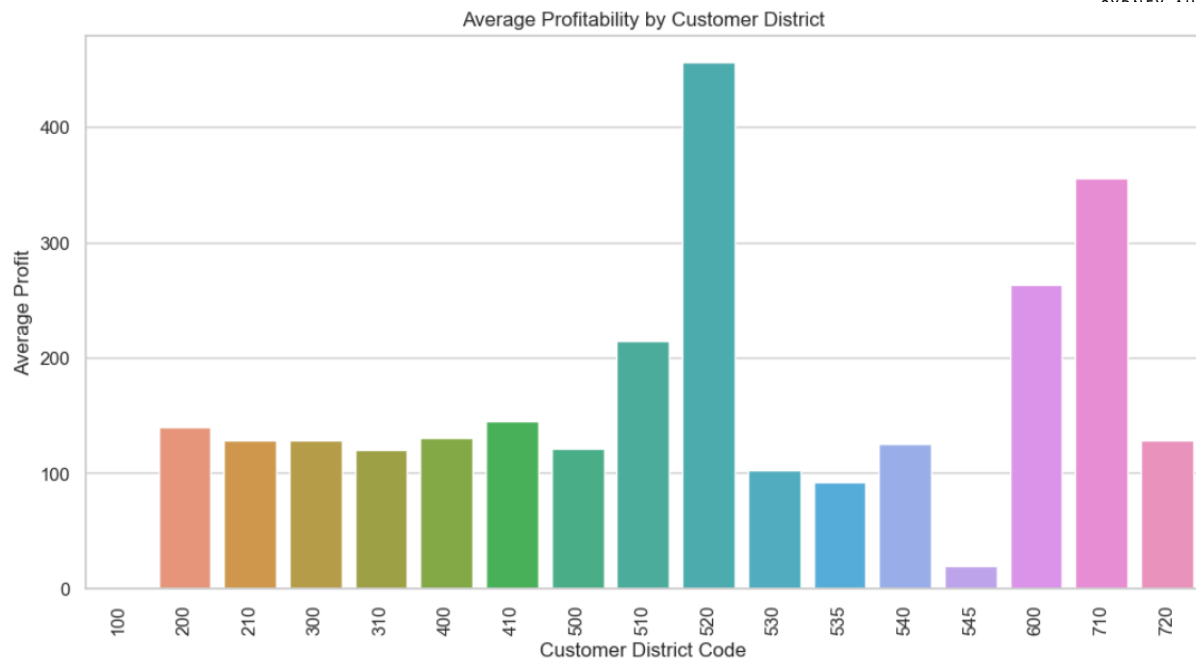
## 2.3 Sales Performance by Hour of Day and Day of Week



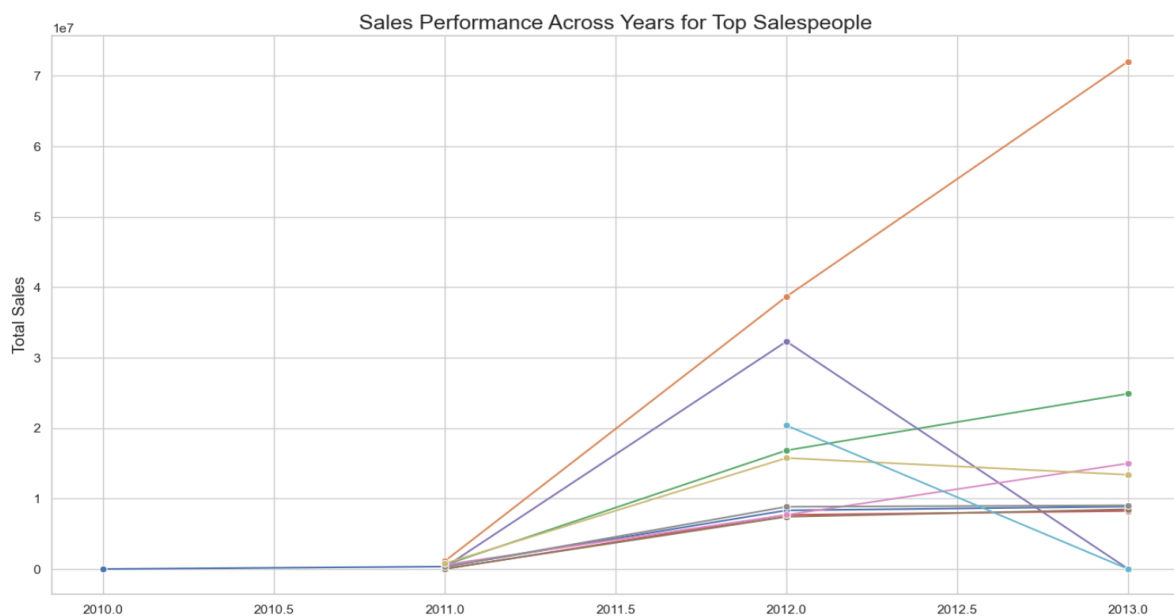
The heatmap visualizes total sales across different days and hours. Sales are highest on weekdays, peaking midweek, with Wednesday and Thursday showing the most activity. Sunday and Saturday see the lowest sales. This pattern suggests focusing marketing and staffing efforts on weekdays, especially around peak hours, to maximize sales performance.

## 2.4 Average Profitability by Customer District

The bar chart titled "Average Profitability by Customer District" shows the average profit for various customer district codes ranging from 100 to 720. The y-axis represents average profit, with values from 0 to 450. District code 520 has the highest average profit, while district code 545 has the lowest. This visualization highlights significant profitability variations across different customer districts, useful for strategic decision-making.



## 2.5 Sales Performance Across Years for Top Salespeople



The line graph "Sales Performance Across Years for Top Salespeople" indicates varied trends: the orange line shows consistent growth, peaking around 70 million units in 2013, while the purple line drops sharply in 2013 after initial growth. The green line shows steady improvement, ending at 30 million units in 2013. These trends suggest focusing on successful strategies, addressing performance dips, and implementing training programs to boost overall sales performance.

## Section 3 – Test Sub Sample Differences

### 3.1 Is there a significant difference in the company's profit margins between the fiscal years 2012 and 2013?

This analysis focuses on understanding if there was a meaningful change in profit margins between the years 2012 and 2013. Here, profit margin is calculated as the difference between `value_sales` and `value_cost`, which represents how much of the revenue remains as profit after costs are covered. By comparing average profit margins across these two years, we aim to see if any shifts in profitability occurred, which could highlight the impact of market conditions or strategic decisions made during this period.

Identifying any significant shifts in profit margins can help management understand whether strategies were effective and whether external conditions affected profitability. It's a way of checking the company's pulse over time to ensure its financial health and resilience.

#### **Methodology:**

We applied a two-sample t-test to compare the average profit margins between 2012 and 2013. This statistical test allows us to determine if the differences in averages are statistically meaningful or if they could be due to random fluctuations.

**Null Hypothesis (H0):** There is no difference in the average profit margin between 2012 and 2013.

**Alternative Hypothesis (H1):** There is a significant difference in the average profit margin between these years.

#### **Results:**

- **T-test Statistic:** -0.8239    **P-value:** 0.4100

Since the p-value is 0.4100, which is much higher than the conventional threshold of 0.05, we do not have enough evidence to reject the null hypothesis. This means there is no statistically significant difference in profit margins between 2012 and 2013.

The results show that the company's profit margins stayed steady between 2012 and 2013, indicating that current strategies effectively maintained profitability.



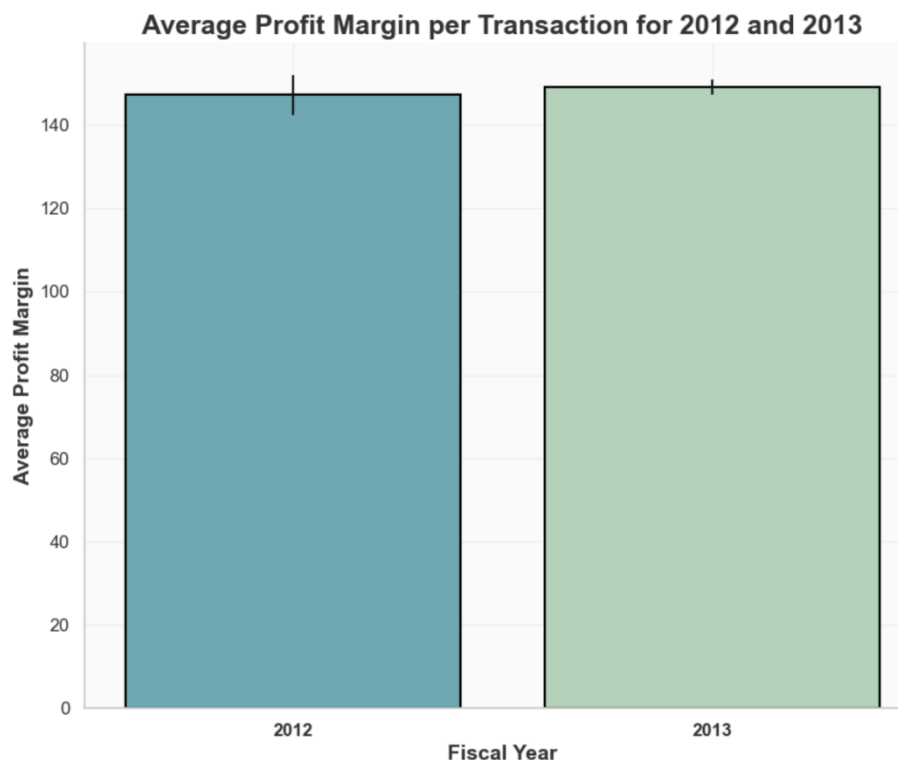
**Effective Strategies:** Stable margins suggest that the company's strategies and operations worked well, even amid potential market shifts. The company can review these successful strategies to reinforce future plans.

**Resilience to Market Changes:** Consistent profitability implies resilience against external changes. The company might explore applying these resilient practices to more volatile or new markets to achieve similar stability.

**Confident Resource Allocation:** With steady margins, management can allocate resources confidently, focusing on areas like product development or market expansion, knowing that core operations are reliable.

**Future Planning and Innovation:** This stability gives a solid foundation to pursue growth initiatives, like launching new products or digital projects, strengthening the company's position without undue financial risk.

In summary, stable profit margins between 2012 and 2013 show that the company's approach is working well. This foundation allows management to confidently plan for growth, explore new opportunities, and stay resilient as the market evolves.





### 3.2 Do Returning Customers Have Higher Order Quantities Than New Customers?

This analysis focuses on whether there's a significant difference in the average order quantities between new and returning customers. Here, new customers are those with an order frequency equal to or below the average, while returning customers order more frequently than average. Knowing if returning customers typically order more can help the company fine-tune its marketing, loyalty programs, and inventory strategies.

By understanding these patterns, the company can decide where to focus its efforts—such as creating incentives to boost order sizes for returning customers or offering promotions to encourage higher first-time orders from new customers.

#### **Methodology:**

We used a two-sample t-test to compare the average order quantities (value\_quantity) for new and returning customers. This statistical test helps us determine if any difference in average order quantities between these groups is likely meaningful or just due to random variation.

**Null Hypothesis (H0):** There is no difference in the average order quantity between new and returning customers.

**Alternative Hypothesis (H1):** There is a significant difference in average order quantities between new and returning customers.

#### **Results:**

Average Order Frequency per Customer: 441.18

- **T-test Statistic:** 18.0374   **P-value:** 1.00e-72

The t-test results (p-value: 1.00e-72) reveal a statistically significant difference in order quantities between new and returning customers. However, contrary to typical assumptions, the results show that **new customers place larger orders on average**. This suggests that new customers may be making bigger initial purchases, possibly due to introductory offers or bulk buying, while returning customers tend to place smaller, more frequent orders.

A loyalty program can sustain engagement with high-value new customers, who typically place larger initial orders, by rewarding repeat purchases. This approach, combined with demand forecasting for new customer acquisition campaigns, helps optimize inventory to meet both initial and repeat order needs.

**Targeted Marketing for Returning Customers:** For returning customers who order smaller quantities, strategies like product bundles, personalized discounts, or "frequent buyer" incentives could encourage larger purchases, increasing overall value from this group.

**Resource Allocation:** This distinction allows the company to better allocate resources—focusing on retaining the value of new customers' initial orders while nurturing returning customers to order more over time.

In summary, the analysis shows that **new customers have higher average order quantities**, guiding strategies for loyalty programs, inventory, and targeted marketing to enhance customer engagement and drive profitability.



## Section 4: Inference

We chose to exclude outliers to make the model more accurate, easier to interpret, and better aligned with statistical assumptions, such as having normally distributed residuals. Outliers can distort key results by skewing the regression line, causing inconsistent error patterns, and masking the real relationships between variables. Removing them reduces the impact of data entry mistakes or rare, unusual events that don't represent day-to-day business activity. This approach creates a more reliable model, clearer visuals, and insights that reflect typical patterns, giving management actionable information they can count on.

### 4.1 Examine how the price adjustments of the items impact the sale value.

#### ***Method: Ordinary Least Squares (OLS)***

OLS regression model is used to examine the correlation between the variables. It will estimate the magnitude and direction of each independent variable in relationship to sales, using `value_sales` as the target. The data is cleaned for non-numeric entries. And the, categorical variables have been encoded in terms of dummy variables for appropriate analysis. Here the column `value_sales` is the dependent variable and the independent variables taken here are `value_price_adjustment`, `fiscal_year`, `customer_district_code` and `item_group_code`.

#### ***Results:***

**R-squared:** 0.170 – This suggests that the model explains around 17% of the variance in `value_sales`. While modest, this indicates that other unobserved factors are also influencing sales.

**Adjusted R-squared:** Also, around 0.170, which shows a consistent explanatory power when adjusting for the number of predictors.

**F-statistic:** 435.2 (p-value < 0.001) – The high F-statistic with a p-value close to zero indicates that the model as a whole is statistically significant, meaning that the chosen independent variables significantly predict `value_sales`.



#### OLS Regression Results

```
=====
Dep. Variable:          value_sales    R-squared:                0.170
Model:                  OLS           Adj. R-squared:           0.170
Method:                 Least Squares  F-statistic:              435.3
Date:                  Wed, 06 Nov 2024 Prob (F-statistic):        0.00
Time:                  00:05:10        Log-Likelihood:           -55569.
No. Observations:      8502           AIC:                     1.111e+05
Df Residuals:          8497           BIC:                     1.112e+05
Df Model:              4
Covariance Type:       nonrobust
=====
```

### ***Evidence of Robustness***

Omnibus and Jarque-Bera Tests: The p-value is near zero, suggesting that residuals are not perfectly normally distributed. This can sometimes indicate model misspecification or the need for transformations on certain variables.

### ***Business Insights for Management***

**Price Sensitivity:** The strong, negative impact of value\_price\_adjustment on sales suggests that customers are sensitive to price adjustments. Management could consider moderating adjustments or implementing strategies to justify these changes (e.g., bundling or adding value-added services).

### ***Geographical Influence:***

The significance of customer\_district\_code highlights the need to explore regional preferences and adjust inventory or marketing efforts to cater to high-performing districts. This insight could help in tailoring promotions or adjusting supply based on regional demand.

### ***Product Grouping:***

Even though item\_group\_code has a minimal effect, its statistical significance suggests that some product groups might consistently outperform others. Management can use this insight to focus on high-performing groups and perhaps rethink strategies for underperforming groups (e.g., product enhancements or targeted marketing).



## 4.2 How do `fiscal_month`, `business_area_code`, and `item_group_code` influence `value_quantity`?

### **Method: OLS**

We use OLS again for the analysis as it helps us easily determine the correlation between the variables. Here the dependent Variable is `value_quantity` the Independent Variables is `fiscal_month`

### **Results:**

R-squared: 0.000 – This very low R-squared value suggests that `fiscal_month` alone explains almost none of the variability in `value_quantity`, indicating that there are likely other variables or factors driving sales quantities.

F-statistic: 245.8 (p-value < 0.001) – Although the R-squared is low, the F-statistic is statistically significant. This indicates that `fiscal_month` has a statistically significant, albeit weak, linear relationship with `value_quantity`.

P-value: < 0.001 – Despite the low R-squared, the `fiscal_month` is statistically significant, indicating a weak but consistent monthly impact on sales quantities.

The box plot of `value_quantity` by `fiscal_month` shows relatively stable distributions across months, with some outliers in certain months. This stability in the box plot supports the regression finding that `fiscal_month` does not have a major impact on `value_quantity`

### **Evidence of Robustness**

Omnibus and Jarque-Bera Tests: The p-value is zero, indicating a departure from normality in residuals, which is typical with high-frequency time series data and could suggest the need for transformation or adding seasonal adjustments.

Durbin-Watson Statistic: 1.232 – This value, close to 1, suggests some autocorrelation in residuals, possibly due to unobserved seasonal effects.

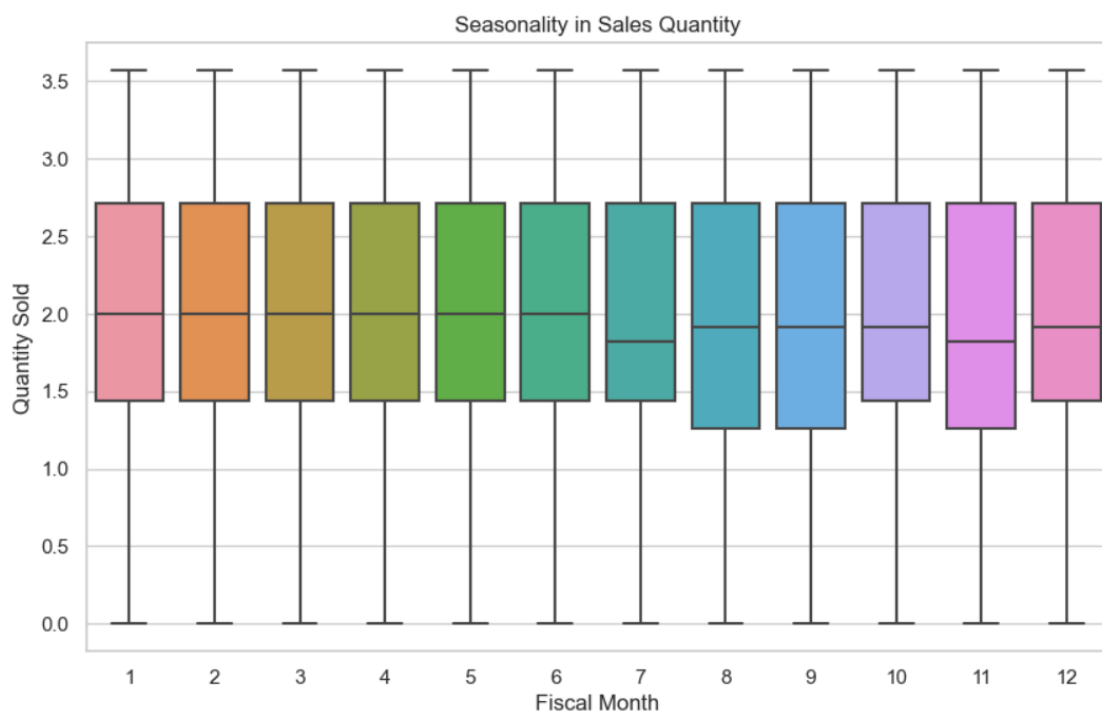
This visualization confirms minor variations across months, which may point to seasonal sales patterns.



### ***Insights for the management***

The results suggest that sales quantity is fairly consistent across months, with minimal seasonality. This insight could imply that inventory levels can remain relatively stable throughout the year, reducing the need for large seasonal adjustments.

This analysis helps the management team plan for potential fluctuations in sales quantity, contributing to more informed inventory management and seasonal sales strategies.



## **Section 5 – Prediction Model**

### **Sales Forecast Report for 2014 Using SARIMA Model**

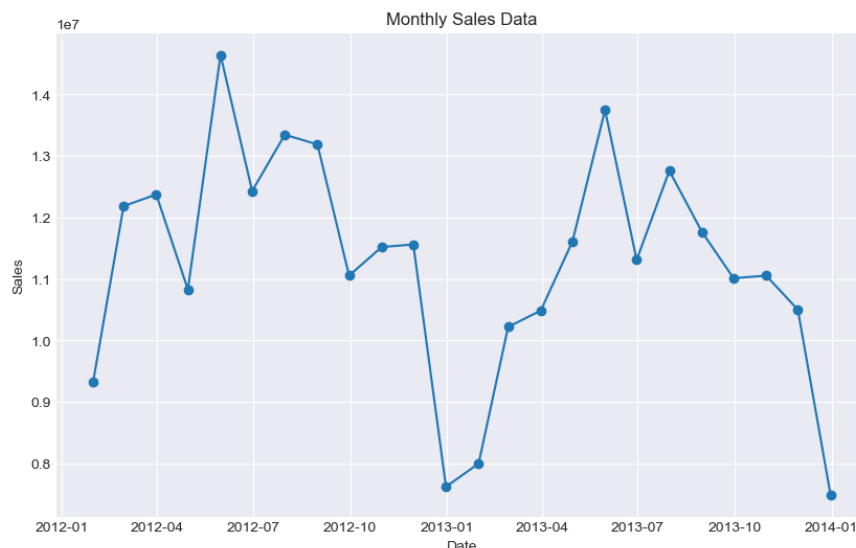
#### **1. Introduction**

The goal of this study is to predict 2014 monthly sales using past data from 2012 and 2013. Due to the distinct seasonal trends seen in past sales data, a Seasonal ARIMA (SARIMA) model was chosen for its ability to account for seasonality in time series data. The goal of the analysis is to offer perspectives on anticipated sales patterns in 2014, aiding in inventory, staffing, and marketing preparation.

## 2. Data Analysis and Preprocessing

- **Data Aggregation:** The dataset consists of monthly aggregated sales data from January 2012 to December 2013, each entry indicating total revenue for a specific month.
- **Outlier Handling:** Outliers were identified and managed using interquartile range (IQR) thresholds to ensure stability.
- **Seasonality:** Initial analysis revealed a recurring seasonal pattern, with peak sales in mid-year and lower sales toward year-end.

A first observation (Figure 1) of the past data showed a repetitive seasonal trend, with highest points in the middle of the year and decreases towards the end of the year. This recurring pattern shows yearly fluctuations in sales that the SARIMA model strives to predict.



**Figure1:** Monthly Sales Data (2012-2013)

- **Insights:** The information indicates steady increases in sales during the middle of the year and a decrease in demand towards the end of each year.

## 3. Model Selection and Parameter Tuning

A SARIMA model was used with a seasonal period of 12 months to capture the annual cycle. A search was done on a grid to discover the best parameters for the model using the Akaike Information Criterion (AIC), which assesses the quality of the model. The best setup was identified as SARIMA (0,1,0)x(0,1,1,12), with an AIC score of 337.23, signaling a strong match for the dataset.

### Parameter Tuning Summary:

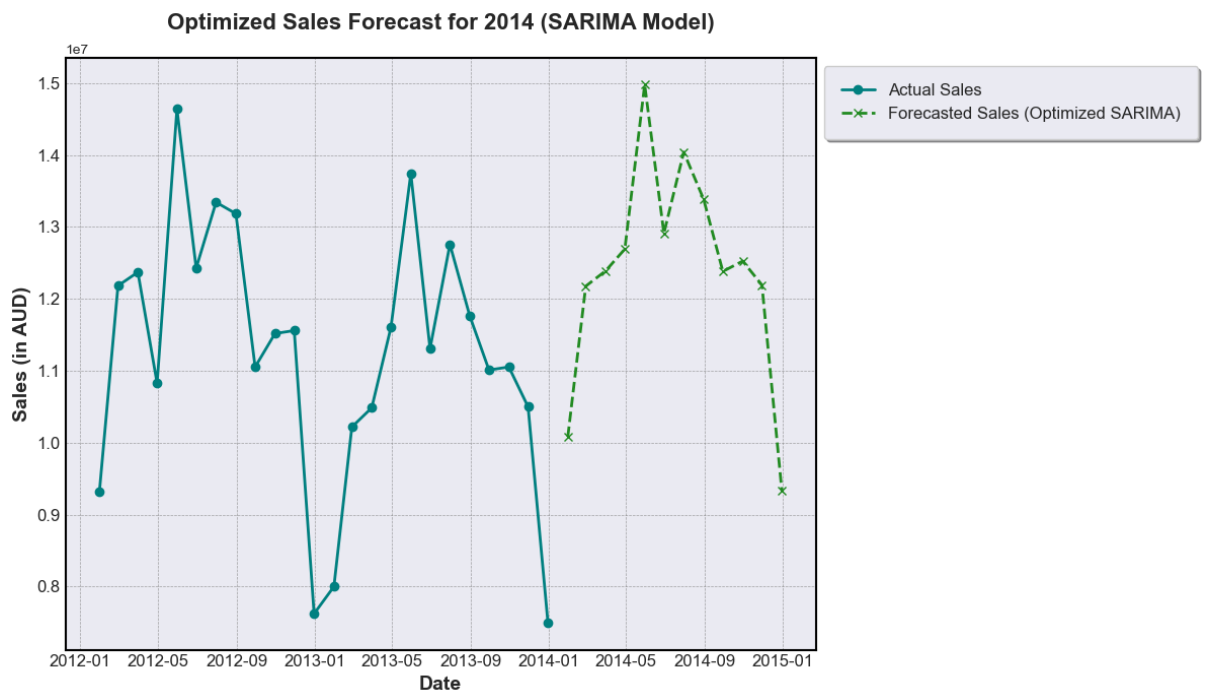
- **Non-Seasonal Order:** (0, 1, 0)
- **Seasonal Order:** (0, 1, 1, 12)
- **AIC:** 337.23

#### 4. Forecast Results

The SARIMA model predicted monthly sales for 2014 by capturing the seasonal trends seen in the past data. As depicted in Figure 2, the prediction suggests comparable high points and low points, with anticipated increased sales in the middle of the year and decreased sales towards the end of the year.

##### Forecast Summary for 2014:

- **Peak Sales:** Mid-year, specifically around May-August.
- **Low Sales:** Beginning and end of the year, particularly in January and December.



**Figure 2:** Optimized SARIMA Sales Forecast for 2014

The forecasted sales values (in AUD) are as follows:

| Month   | Forecasted Sales | Month | Forecasted Sales |
|---------|------------------|-------|------------------|
| January | 10.08M           | July  | 15.26M           |



|                 |        |                  |        |
|-----------------|--------|------------------|--------|
| <b>February</b> | 12.56M | <b>August</b>    | 14.85M |
| <b>March</b>    | 13.02M | <b>September</b> | 13.96M |
| <b>April</b>    | 12.43M | <b>October</b>   | 13.59M |
| <b>May</b>      | 14.87M | <b>November</b>  | 12.78M |
| <b>June</b>     | 14.98M | <b>December</b>  | 9.34M  |

These forecasts align with historical trends, providing a realistic outlook for 2014 based on past seasonality.

## 5. Seasonal Patterns and Business Insights

The SARIMA model accurately captures the seasonal pattern of rising demand in the middle of the year and declining sales towards the end of the year. These observations enable the company to:

- **Inventory Planning:** Raise inventory levels prior to the expected mid-year peak in order to satisfy increased demand.
- **Staffing Changes:** Increase manpower during peak months for better customer service.
- **Marketing Strategies:** Concentrate marketing activities on mid-year months, which usually see increased sales.

Comprehending these trends can assist the company in making data-informed choices to improve operational efficiency and customer satisfaction.

## 6. Conclusion and Recommendations

The SARIMA model has offered a dependable prediction that mirrors the seasonal sales trend from the past. The knowledge acquired is helpful for planning and is anticipated to aid in enhancing inventory management, staffing, and marketing tactics.

### Recommendations for Strategic Planning Based on Sales Forecast:

#### 1. Inventory Management:

- **Stock Up for Peak Demand:** Increase inventory before mid-year peaks to avoid stockouts.



- **Reduce Inventory for Low-Demand Periods:** Lower stock levels at year-end to minimize holding costs.

## **2. Staffing Adjustments:**

- **Boost Workforce During Peak Months:** Increase staffing in mid-year to handle higher demand.
- **Scale Back During Slow Periods:** Adjust staffing in low-demand months to save costs.

## **3. Marketing Strategies:**

- **Intensify Marketing During Peak:** Run promotions mid-year to capture more demand.
- **Off-Season Discounts:** Use discounts at year-end to stimulate sales during slower months.

## **4. Financial and Operational Planning:**

- **Align Budget with Demand:** Allocate more resources for peak periods and plan cash flow accordingly.
- **Update Forecast Regularly:** Refresh the SARIMA model with new data annually for improved accuracy.



## Section 6: Higher Likelihood of Losing Customers

### Summary

Outliers play a critical role in the analysis of customer churn as they can significantly influence the results and insights derived from the dataset. By identifying and addressing outliers, we ensure that the analysis reflects true customer behaviour rather than being skewed by atypical values. This is essential for achieving a clear and accurate understanding of customer trends, allowing us to make informed decisions based on reliable data. Incorporating outliers in our analysis enables us to recognize extreme cases that may indicate underlying issues or opportunities, ultimately enhancing the overall effectiveness of our churn analysis.

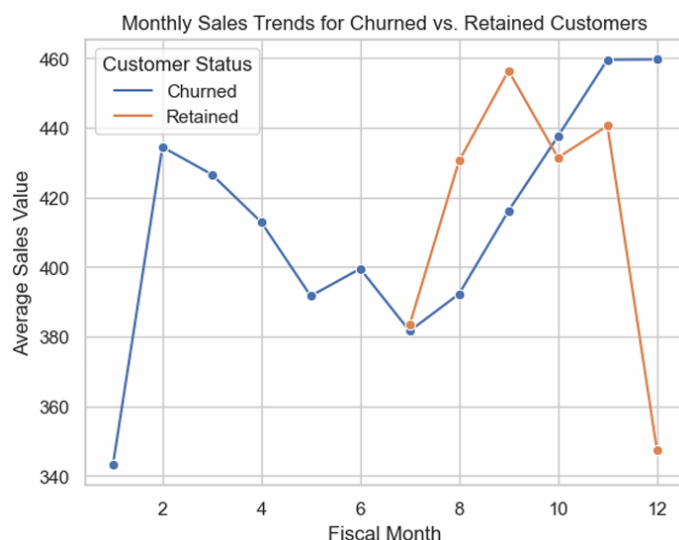
Businesses must comprehend customer turnover since it has a direct effect on profitability and client retention. The analysis seeks to find trends and churn-causing causes using clean and trustworthy data, enabling the creation of focused retention tactics.

The report's insights will help guide proactive steps to boost overall business performance and increase client loyalty. Putting these strategies into practice will be essential to building enduring connections with customers and maximizing revenue.

### Analysis and Key Insights

#### Monthly Sales Trends for Churned vs. Retained Customers

An analysis of monthly sales trends reveals a significant decline in sales for churned customers during months 5 to 8, whereas retained customers exhibit more stable or increasing trends throughout the year. Churned customers experience a sharp drop in sales in month 12, indicating that many may cease purchasing entirely by the end of the fiscal year. This mid-year decline serves as an early warning sign for potential churn, offering the business a window to intervene.



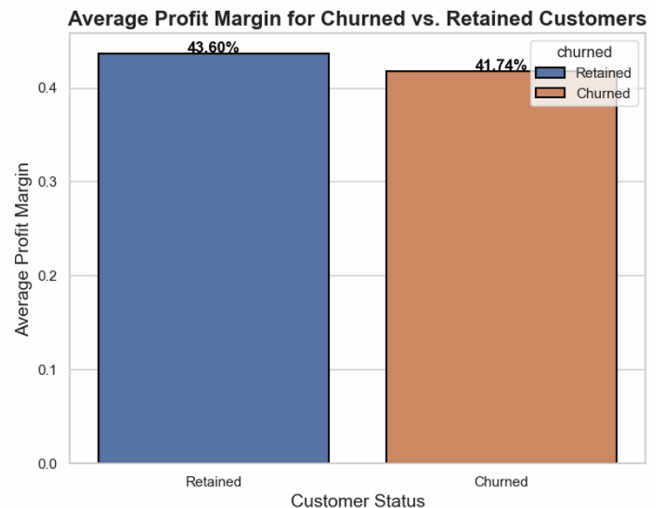
**Actionable Recommendation:** The business should implement re-engagement campaigns targeting customers with reduced sales activity during months 5 to 8. By offering personalized



promotions or loyalty rewards, the business may be able to sustain engagement and reduce the risk of churn.

## Profit Margins for Churned vs. Retained Customers

Retained customers demonstrate slightly higher average profit margins (43.6%) compared to churned customers (41.74%). While the difference is small, it underscores the financial value of retaining customers who are more profitable on average. This finding supports the importance of focusing retention efforts on customers who not only provide consistent revenue but also contribute positively to the company's profit margins.



**Actionable Recommendation:** Strengthening loyalty incentives, such as exclusive offers and personalized discounts, could help retain these high-value customers. By focusing on enhancing customer satisfaction and encouraging repeat purchases, the business can sustain its profitability while minimizing churn.

## Regional Churn Variability

A regional analysis of customer churn shows a clear disparity in churn rates across different customer districts. Certain districts, such as 200 and 210, have notably high churn rates (around 78% or more), while other districts, like



530 and 540, show relatively lower churn rates (60-70%). This variability suggests that customer churn may be influenced by localized factors, such as competition, service quality, or regional market dynamics. Understanding these factors can enable the business to tailor its retention strategies more effectively.

**Actionable Recommendation:** The business should conduct a deeper investigation into high-churn districts to understand potential causes, such as local service gaps or competitive



pressures. Implementing district-specific marketing and retention efforts could help address the unique needs of customers in high-churn areas, potentially improving retention. In contrast, lower-churn districts can be analysed for best practices, which can then be applied to other regions facing higher churn rates.

## Correlation Analysis of Features with Churn

A correlation heatmap was used to examine the relationship between key features (e.g., value\_sales, value\_quantity, profit\_margin, and value\_price\_adjustment) and churn. The analysis reveals weak correlations between these features and churn, with the highest correlation observed between value\_sales and value\_quantity, which is expected but not strongly related to churn prediction. This lack of strong correlation suggests that churn may not be directly influenced by these basic transactional features alone.



**Actionable Recommendation:** To gain a deeper understanding of churn drivers, the business should expand data collection to include metrics such as customer satisfaction scores, engagement frequency, and service interaction records. Additionally, employing machine learning techniques could uncover non-linear relationships or complex interactions that are not evident through simple correlation analysis. This approach would improve churn prediction accuracy and enable more targeted retention efforts.

## Key Churn Indicators and Recommendations

Based on the insights gathered from the analysis, the following table summarizes the main churn indicators and corresponding recommended actions:



| Churn Indicator                     | Insight                                                    | Action                                                                         |
|-------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------------|
| Mid-Year Sales Decline              | Sales drop in months 5-8 for churned customers.            | Re-engage with at-risk customers during this period through targeted offers.   |
| Regional Churn Variability          | High churn rates in districts 200, 210, etc.               | Implement region-specific strategies to address local retention challenges.    |
| Higher Profitability of Retained    | Retained customers have higher profit margins on average.  | Focus on loyalty programs to keep high-value customers engaged and satisfied.  |
| Weak Correlation with Analysed Data | Basic features do not strongly predict churn on their own. | Collect more data on satisfaction and engagement and use predictive modelling. |

## Recommendations

- 1. Re-engage At-Risk Customers Mid-Year:** Customers whose sales decline during months 5 to 8 should be the focus of targeted re-engagement campaigns. Personalized offers, loyalty rewards, or outreach during this period can help to maintain customer engagement and reduce churn.
- 2. Localized Retention Strategies:** Regional differences in churn rates indicate a need for district-specific retention strategies. In high-churn areas, the business should address potential local challenges by tailoring marketing efforts, improving service quality, or adjusting pricing strategies.
- 3. Enhance Loyalty Programs for High-Value Customers:** Since retained customers show slightly higher profit margins, implementing loyalty programs that reward high-value customers can sustain this profitability. Exclusive discounts, tailored recommendations, and priority support are examples of initiatives that can enhance loyalty.
- 4. Expand Data Collection and Leverage Predictive Analytics:** To gain a comprehensive view of churn drivers, the business should collect additional data on customer satisfaction, engagement, and service interactions. Advanced analytics,



including machine learning, could then be used to detect patterns and improve churn prediction accuracy.

The customer churn analysis highlights mid-year sales declines, regional churn variability, and slight profitability differences between retained and churned customers as key indicators. To enhance retention, the business should focus on re-engagement efforts during the mid-year period, implement district-specific retention strategies, and incentivize loyalty among high-value customers. Expanding data collection and applying predictive modelling techniques will further refine churn prediction capabilities, allowing for proactive and targeted retention strategies. Implementing these recommendations will help the business improve customer loyalty, reduce churn, and achieve sustainable profitability.

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## **CONCLUSION**

This report covers the sales and customer data LuminaTech has from 2012 and 2013. During the EDA, we learned that data cleaning and standardization were extremely important to maintain the integrity of the dataset and more importantly to be able to derive meaningful insights from it. Looking closely at the data from 2012 to 2013, we saw that LuminaTech managed to keep profit margins steady, showing that their strategies held up well despite market shifts. By breaking down sales performance by region and time, we also uncovered important differences across customer districts, which can guide more focused marketing and staffing efforts. These insights allow LuminaTech to make smarter resource decisions, like concentrating efforts on high-performing areas and adjusting operations to match regional demand patterns..

One of the major takeaways from our analysis was the difference in new vs returning customers. And contrary to popular belief, new customers spend larger orders on average than returning customers. Such a discovery is crucial for the design of loyalty programs and marketing promotion, since it implies that the priority of marketing adaptation should be given to new-start customers who are encouraged to be converted to the second-order customers together with the re-buyers encouraged to increase the order size.

The SARIMA model useful forecasts for 2014, which sales peaks and lows were to be expected. Such knowledge allows LuminaTech to respond proactively by increasing or decreasing inventory levels and staffing resources in accordance with forecast demand, thereby increasing operational efficiencies and consumer satisfaction. Grasping how these seasonal trends work is key to making the best of the marketing efforts, and keeping up the customer expectations with the stock availability over the year.

In addition, we highlighted customer churn risks — emphasizing the impact of declines in mid-year sales and churn behaviour variation across regions. Analysis of common dates in which order placements fail to re-engage during times of high-risk while zoning in on distinct districts in retaining customers at a local, targeted, level, can optimize customer retention and minimize averted revenue loss through LuminaTech.

Using these findings helps LuminaTech accomplish the following strategies to take advantage of these insights: improving loyalty practices for most valuable customers, higher level retention initiatives at the district-level, and performing additional collection to better understand customer behaviour. Besides, advanced predictive analytics can be employed to further enhance churn prediction capabilities, enabling proactive and concern-based retention strategies.

Overall, this report not only emphasizes the value of analytics in improving business processes but also the fact that a data-driven approach to influencing strategic decisions is immensely beneficial in the success of an organization. Leveraging the insights gained from this analysis helps LuminaTech to maintain its competitive edge, profitability, and growth which is sustained over a long-term in a dynamic market environment.